Protocol v3.1

Two-party vertical federated regression with threshold decryption
and lattice DP releases, under the honest-but-curious model

Draft for review

Abstract

A researcher R holds a cohort with features $\mathbf{X}_R$. An organization O holds a register with features $\mathbf{X}_O$ and the response $\mathbf{y}$. Protocol v3.1 computes a ridge-regularized least-squares fit over the records they share, together with confidence intervals. It reveals nothing to O beyond public sizes, and it gives R only differentially private releases of the statistics that depend on O's data.

v3.1 revises v3 to fix the four critical issues raised by the independent review of v3, under the honest-but-curious model. It also includes the review's other fixes that apply under this model. The key changes are four:

- O's decryption share carries the DP noise, and the ciphertext component c_0' is never transmitted. This closes a key-share leak.
- The homomorphic input checks are removed. They addressed only a malicious researcher, and one of them could be bypassed.
- Every error term that depends on O's data or secrets is flooded statistically, in ℓ_2, by the DP noise itself.
- The ridge is chosen from a separate first release, so it is independent of the noise in the release used for estimation.

Security against a deviating researcher is stated as an extension that requires zero-knowledge input proofs.

1 Setting, threat model and guarantees

Data. O restricts its register to a study scope $D_\Phi = \{j : \Phi(j)\}$, where the predicate Φ uses public attributes only, such as province and birth-year band. O pads D_Φ with dummy rows (fresh random identifiers that never match, with $\mathbf{x}_O = \mathbf{0}$ and $y = 0$) to a public size $\bar{n}_O$. R holds $n_R \le n_R^{\max}$ cohort records. Let I be the cohort members that appear in D_Φ, and $n = |I|$. The target is the ridge fit on $\mathbf{G} = \mathbf{X}^{I\top}\mathbf{X}^I$ and $\mathbf{c} = \mathbf{X}^{I\top}\mathbf{y}^I$ over $\mathbf{X}^I = [\mathbf{X}_R^I \mid \mathbf{X}_O^I]$, with $p = d_R + 1 + d_O$ columns, where the $+1$ is R's intercept. In block form,

$$\mathbf{G} = \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{B} \end{pmatrix}, \qquad \mathbf{c} = \begin{pmatrix} \mathbf{c}_R \\ \mathbf{c}_O \end{pmatrix},$$

where $\mathbf{A} = \mathbf{X}_R^{I\top}\mathbf{X}_R^I$ is R's own block, and $\mathbf{B}, \mathbf{C}, \mathbf{c}_R, \mathbf{c}_O, \mathbf{y}^\top\mathbf{y}$ depend on O's data.

Preprocessing. Before the protocol, each party transforms its own columns with constants agreed in Setup from public reference data or domain knowledge (Preprocess, Algorithm 7): it standardizes each column; it may apply a per-column tail transform (winsorizing at $|z| \le c$, a rank-Gaussian transform from public reference quantiles, or a log transform); and it may normalize

its own block of every row to unit norm (R without the intercept, so $\|\dot{\mathbf{x}}\|^2 = 2$; O gets $\|\mathbf{x}_O\| = 1$), optionally appending a clipped log of the block norm as a feature. O treats y the same way, except for block normalization. No transform uses the other party's columns, so no joint computation is needed. Each transform is a fixed function of one record and public constants, so one changed raw record changes one transformed record, and the adjacency and sensitivity analysis apply unchanged to the transformed data, with the bounds chosen on the transformed reference data. O's constants must be data-independent, like CALIBRATE's outputs; R may use its own constants, but it commits to B_R. Block normalization changes the meaning of the coefficients and is intended for prediction; winsorizing and log transforms keep them interpretable. Joint normalization of whole rows is not used: it would make $\mathbf{A}$ depend on O's data and need nonlinear functions under encryption, and on real data a joint norm budget gave little or no gain, because heavy tails, not the lack of a joint bound, make the bounds loose.

Threat model. Both parties are *honest-but-curious*. They follow the protocol, including key generation and R's preparation of its inputs (clipping, the selection vector), and they try to learn what they can from their views. Channels are private and authenticated. R is the analyst and receives the output.

Definition 1 (Datasets and adjacency)**.** A dataset is O's padded scope register $D = \{(\mathrm{id}_j, \mathbf{x}_{O,j}, y_j)\}_{j \le \bar{n}_O}$, with the cohort $(\mathrm{id}, \mathbf{X}_R)$, the identifiers and Φ held fixed. $D \sim D'$ if they differ in $(\mathbf{x}_O, y)$ of exactly one row, matched or not.

Definition 2 (SIM-CDP and IND-CDP, after Mironov, Pandey, Reingold and Vadhan, 2009)**.** A protocol is (ε, δ)-SIM-CDP for R with statistical slack γ if there exist a mechanism $\mathcal{M}$ that is (ε, δ)-DP with respect to $\sim$ and a probabilistic polynomial-time simulator $\mathcal{S}$ such that, for every D and every PPT distinguisher $\mathcal{A}$, $|\Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D)) = 1] - \Pr[\mathcal{A}(\mathcal{S}(\mathcal{M}(D))) = 1]| \le \gamma + \mathrm{negl}(\lambda_{\mathrm{sec}})$. It is (ε, δ)-IND-CDP if $\Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D)) = 1] \le e^{\varepsilon} \Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D')) = 1] + \delta + \mathrm{negl}(\lambda_{\mathrm{sec}})$ for all $D \sim D'$. SIM-CDP with slack γ implies IND-CDP with $\delta + (1 + e^{\varepsilon})\gamma$.

	Guarantee	Basis
G1	**R-input privacy**: O's view can be simulated from $(n_R^{\max}, \bar{n}_O)$ and the public parameters.	RLWE in s_R; security of the PSI. O never receives a decryption share from R.
G2	**Cohort privacy**: O learns nothing about R's cohort beyond $n_R^{\max}$.	Padded OPRF queries; labeled PSI (Chen et al., 2018).
G3	**Output privacy**: per run, $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP with slack γ, hence $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$-IND-CDP; over all runs on a base register, $(\varepsilon_L, \delta_L)$-IND-CDP.	Theorem 1; Proposition 1.
G4	**Disclosed**: R learns I (which cohort members are in D_Φ) and uniformly random positions $\tau(j)$, and nothing else about O's register.	By design (Mode L; Remark 2); OPRF-keyed items and labels; SANITIZEBFV.

Table 1: Guarantees of v3.1 against honest-but-curious parties. Correctness assumes honest computation.

2 What changes relative to v3

Other review items that apply under the honest-but-curious model.

Critical issue in v3	Exploitable by	Fix in v3.1
C1: R shifts its decryption share to pass the input check	malicious R	No homomorphic input check. Under the honest-but-curious model R prepares $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ as specified (Algorithm 3). The check, the slack vector and the relinearization and conjugation keys are all removed. Robustness to a deviating R is the extension in Section 8.
C2: inputs that wrap around modulo q pass the check	malicious R	As for C1. In addition, q_0 is sized so that honest inputs never wrap (Algorithm 7, CALIBRATE).
C3: the release step leaks O's key share	honest-but-curious R	O sends (c_1', h_O) with $h_O = c_0' + c_1' s_O + \eta$; c_0' is never transmitted. Fresh (u, e_0, e_1, η) for every release. The stages use separate packed ciphertexts, with no masking of a larger one (Algorithms 4–5).
C4: the check tolerance is not carried into the sensitivity	malicious R	With no approximate check there is no tolerance. B_R is defined to include the intercept, so honest rows satisfy the bound exactly.

Table 2: The four critical issues of the review and how v3.1 resolves them.

- **Flooding.** The DP noise statistically floods every error term in the decrypted value that depends on O's data or secrets. These are: the error of the evaluated circuit, including key-switching error with O's key-share errors; the encoding and rescaling roundings; and the fresh-encryption error. The bound is in ℓ_2 over the whole ring element, worst case over O's data (Section 5).
- **Exact noise.** An exact big-integer discrete Gaussian sampler. Floating point is forbidden, because at $\sigma_c \approx 2^{52}$ it leaves the low bits un-noised.
- **Scope.** Φ uses public attributes only, and D_Φ is padded to a public $\bar{n}_O$.
- **Linkage.** A fresh OPRF key and permutation for every run; OPRF queries padded to $n_R^{\max}$; a cross-session OPRF quota per researcher and project.
- **Ledger.** The ledger is kept per base register and shared by all researchers. It charges a run's full cost, for all planned releases, when it admits the run, atomically and without refund, and it accounts the smudging step in hockey-stick divergence. δ_{PSI} is a correctness event and is not charged to the DP δ.
- **Statistics.**
  - An explicit concentration radius for the stage-1 lower bound.
  - The ridge (λ, Ψ), with $\Psi \in \{\Pi_O, \mathbf{I}\}$, chosen from stage 1 and $\mathbf{A}$ only.
  - Two clearly defined estimands.
  - A parametric bootstrap for the mechanism variance and for the RSS correction.
  - A public a-priori bound on $\|\boldsymbol{\beta}\|$ when the private bound is infinite.
  - An explicit split of α.
- **Upload.** Seeded symmetric encryption under s_R, converted by O, halves the upload.

Major issues 1 (malicious key generation), 6 (enforcing the study scope) and 7 (resolution of the input check) apply only to a deviating party. They are covered in Section 8.

3 Notation and public parameters

Parameters. The protocol is stated for a parameter vector $\mathcal{P}$ with explicit admissibility constraints; Section 7 lists it. CALIBRATE derives the flooding parameter, the noise, the scales and the size of q_0

Symbol	Meaning
Φ, $\bar{n}_O$, $n_R^{\max}$	study scope (public attributes); padded register size; certified cohort cap
B_R, B_O, B_y	clipping bounds on the preprocessed data: $\|\dot{\mathbf{x}}\| \le B_R$ for R's full row *including* the intercept; $\|\mathbf{x}_O\| \le B_O$; $\|y\| \le B_y$
r, o, w	B_R^2, B_O^2, B_y^2
$\Delta, \Delta_{\neg yy}, \Delta_G, \Delta_y$	replace-one ℓ_2 sensitivities in the slot domain: full release with and without $\mathbf{y}^\top\mathbf{y}$; (triu $\mathbf{B}, \mathbf{C}$) only; $\mathbf{y}^\top\mathbf{y}$ alone ($= w$)
$(\varepsilon_{\text{run}}, \delta_{\text{run}})$	per-run budget; f_1 = stage-1 share; f_y = share for a separate $\mathbf{y}^\top\mathbf{y}$ release (parameters, Section 7)
ρ_s	zCDP cost of release s; $\rho_{\text{run}} = \sum_s \rho_s$
γ, δ_{DP}, ζ	statistical slack of a run; the part of δ_{run} left to the DP mechanism; largest share of δ_{run} the slack may take
$(\varepsilon_L, \delta_L)$, ζ_L, ξ_L, δ'; ρ_L, E'_L, Γ_L	ledger cap of a base register, its slack share, its share reserved for smudging and the per-release slack of smudging; derived zCDP, smudging and slack capacities
$\vartheta, \vartheta^\star$	validity ratio $\lambda_{\min}(\hat{\mathbf{G}} + \lambda\Psi)/(2\sigma_2\sqrt{p})$; its target
N, $N_s = N/2$, q_0	ring dimension, slots, last-level modulus
$\Omega_U, \Omega_u, \Omega_V, \Omega_f$	encoding scales of R's vectors, of the masks and of O's data; exact final scale $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1q_2)$
$\mathsf{Ecd}, \mathsf{Dcd}$	encoding $\mathsf{Ecd}_\Omega(\mathbf{z}) = \lfloor \Omega\, \tau^{-1}(\mathbf{z}) \rceil$ and decoding $\mathsf{Dcd}(v) = \tau(v)/\Omega_f$, where τ is the canonical embedding
$\sigma_{c,s}$, σ_s	noise scale on the coefficient lattice, and the induced slot-domain s.d. $\sigma_s = \sigma_{c,s}\sqrt{N/2}/\Omega_f$
$B_{\text{tot},s}$, δ_B, κ	ℓ_2 bound on all flooded error terms, holding with probability $\ge 1 - \delta_B$; flooding parameter (distance $\le 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ per release), derived by CALIBRATE
ξ_T, T_s	failure bound of one call of the budgeted sampler (one release, or one flooding vector of SANITIZEBFV); its fixed number of candidate trials for release s, from $(\sigma_{c,s}^2, N, \xi_T)$
$\mathcal{E}$, $\mathcal{E}_G$, $\mathcal{E}_2$	released entries: all; (triu $\mathbf{B}, \mathbf{C}$); stage-2 entries ($\mathbf{y}^\top\mathbf{y}$ included only in joint mode)

Table 3: Notation. ρ always denotes a zCDP budget and Δ always a sensitivity; CKKS scales are Ω.

from $\mathcal{P}$, or refuses the run.

4 Protocol

Figure 1 shows the message flow. Algorithms 1–7 give the details.

5 The release mechanism and output privacy

Encoded statistic and centre. For release s, let $\mathbf{z}_s(D) \in \mathbb{R}^{N_s}$ hold the released entries in their slots and zeros elsewhere. The mechanism is centred at the noiseless circuit output $f_s(D) = \lfloor \hat{\mathbf{z}}_s(D) \rceil \in \mathbb{Z}^N$, with $\hat{\mathbf{z}}_s(D) = \sum_{e \in \mathcal{E}_s} (q_1q_2)^{-1} \mathrm{Tr}\big(\sum_h m_{U_e,h} m_{V_e,h}\big) m_{u_e}$, where $m_{U_e,h}$, $m_{V_e,h}$, m_{u_e} are the encodings at Ω_U, Ω_V, Ω_u. f_s depends on the data and public parameters only. It differs from $\mathsf{Ecd}_{\Omega_f}(\mathbf{z}_s(D))$ by deterministic terms, chiefly the mask spill, which are part of the query and need no flooding.

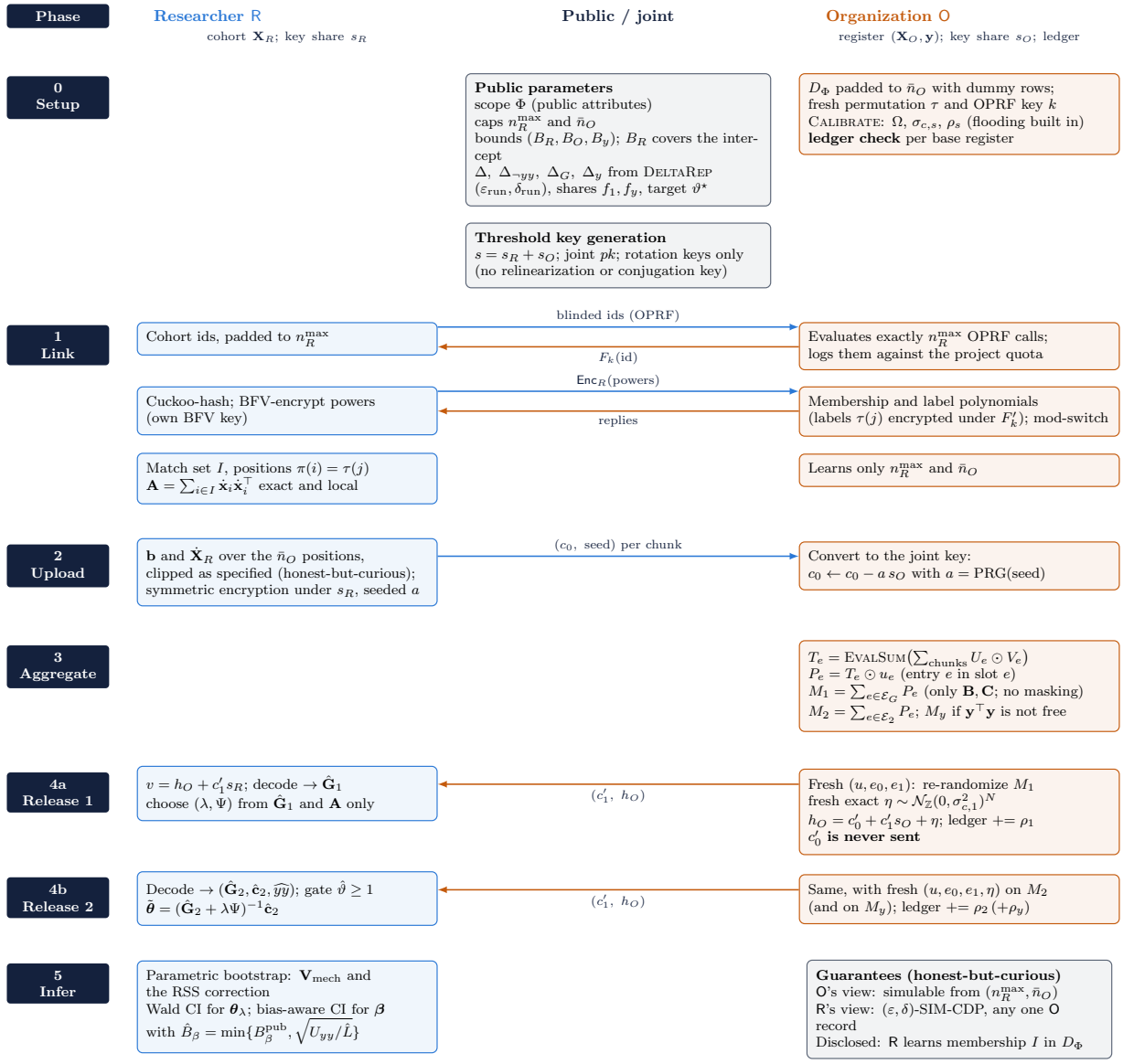


Figure 1: Message flow of protocol v3.1. Blue arrows go from R to O and orange arrows from O to R. O never receives a decryption share, and R never receives c'_0.

Algorithm 1 Setup

Both: agree on the parameter vector $\mathcal{P}$ (Table 4): Φ (public attributes only), $n_R^{\max}$, $\bar{n}_O$, $(\varepsilon_{\text{run}}, \delta_{\text{run}})$, the ledger parameters $(\varepsilon_L, \delta_L, \zeta_L, \xi_L, \delta')$, f_1 (f_y), ζ, $\kappa_{\min}$ (or a fixed κ), δ_B, t_η, $\vartheta^\star$, $\alpha_{\text{CI}}, \alpha_L, \alpha_U$, B_β^{pub}, $\bar{\sigma}^2$, κ_Ψ, η_B, B, the PSI parameters, N, the chain, Ω_U, Ω_u

O: $D_\Phi \leftarrow \{j : \Phi(j)\}$; pad with dummy rows to $\bar{n}_O$; draw a fresh secret permutation τ of $[\bar{n}_O]$

R: and **O:** , each on its own columns only: PREPROCESS with constants from public reference data or domain knowledge

Both: $(B_R, B_O, B_y) \leftarrow$ CHOOSEBOUNDS on the preprocessed reference data, or from domain knowledge ▷ B_R covers the intercept

$(\Delta, \Delta_{\neg yy}) \leftarrow \text{DELTAREP}(B_R, B_O, B_y)$; $\Delta_G \leftarrow \text{DELTAREP}(B_R, B_O, 0)$; $\Delta_y \leftarrow B_y^2$

mode $\leftarrow$ joint if $\Delta - \Delta_{\neg yy} \le 10^{-9}\Delta$, otherwise separate ▷ release $\mathbf{y}^\top\mathbf{y}$ jointly only when it is free

O: $(\kappa, \Omega_V, \Omega_f, \{\sigma_{c,s}, \rho_s\}_s, \mathit{cost}) \leftarrow \text{CALIBRATE}(\mathcal{P})$; refuse if it refuses ▷ Algorithm 7; R repeats it

O: atomically: **if** the ledger of the base register admits the run's cost $(\rho_{\text{run}}, \varepsilon'_{\text{run}}, \gamma_L)$ (Section 5) **then** charge it in full **else** refuse ▷ no refund

Both: threshold key generation (Mouchet et al., 2021) with a common random a: $s = s_R + s_O$, $pk = (p_0, a)$ with $p_0 = -as + e$, and rotation keys for $\{2^i\}$
(no relinearization or conjugation key)

O: fresh OPRF key k; per-run PSI preprocessing on $\{F_k(\text{id}_j)\}$ with labels $\tau(j)$ encrypted under $H(F'_k(\text{id}_j))$
padding (bin degree, number of partitions) public; log the run against the quota for (researcher, project)

Lemma 1 (Sensitivity of the encoded statistic and of the centre)**.** $\|\tau^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\,\|\mathbf{z}\|_2$, *so* $\sup_{D\sim D'} \|\mathsf{Ecd}_\Omega(\mathbf{z}_s(D)) - \mathsf{Ecd}_\Omega(\mathbf{z}_s(D'))\|_2 \le \sqrt{2/N}\,\Omega\,\Delta_s + \sqrt{N}$. *For the centre (proofs, Part A),*

$$\sup_{D\sim D'} \|f_s(D) - f_s(D')\|_2 \le \widehat{\Delta}_{\text{enc},s} := \sqrt{2/N}\,\Omega_f\Delta_s + \Delta_{x,s} + \sqrt{N},$$

with $\Delta_{x,s}$ *an explicit function of* N*, the scales,* q_1, q_2*,* $|\mathcal{E}_s|$*, the bounds and* Δ_s*.* Ω_f *is the exact final scale, tracked exactly;* R *decodes with it.*

Error decomposition. Write the value R reconstructs in Algorithm 5 as

$$v_s = h_O + c'_1 s_R = c'_0 + c'_1 s + \eta = f_s(D) + e_{\text{tot},s} + \eta_s \pmod{q_0}.$$

$e_{\text{tot},s} \in \mathbb{Z}^N$ collects every term that depends on a key, an error or a coin: R's encryption errors times O's data, the rescaling and ModDown roundings, the key-switching error (which involves O's key-share errors), the mask spill of these errors, the rounding of the centre, and the fresh-encryption error $e\,u + e_0 + e_1 s$.

Obligation 1 (Worst-case error bound; discharged in the proofs, Part A, Theorem O1)**.** For the circuit of Algorithm 4, $B_{\text{tot},s}$ is an explicit function of public parameters only (N, $\bar{n}_O$, d_R, d_O, the bounds, the primes and the key-switching gadget, $\Omega_U, \Omega_V, \Omega_u$, δ_B, the error distribution) with $\Pr[\|e_{\text{tot},s}\|_2 \le B_{\text{tot},s}] \ge 1 - \delta_B$ for every dataset D in the clipped domain, the probability being over the keys and fresh randomness only. CALIBRATE must use this worst-case bound, never a value estimated from the ciphertexts.

Lemma 2 (Smudging)**.** *If* $\eta \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_c^2)^N$*,* $\sigma_c \ge \max\{2^\kappa B_{\text{tot}}/\sqrt{2\pi}, 2^{42}\}$ *and* $\|e\|_2 \le B_{\text{tot}}$*, then* $\text{SD}(\eta + e, \eta) \le 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$*, also given side information independent of* η*.*

Algorithm 2 Link (labeled PSI with an OPRF)

R: pad the cohort identifiers with random dummies to exactly $n_R^{\max}$
R: run the OPRF with O on each padded identifier $\to F_k(\mathrm{id})$ ▷ O evaluates exactly $n_R^{\max}$ calls
R: cuckoo-hash the PRF values; send BFV encryptions of windowed powers under R's own BFV key
O: evaluate the membership and label polynomials per bin and partition; mask; SanitizeBFV; reply ▷ flooding κ_ψ; abort if a sampler call fails
R: decrypt $\to$ the match set I and positions $\pi(i) = \tau(j)$ for $i \in I$ (labels decrypted with keys from $F'_k(\mathrm{id})$); abort on an ambiguous or out-of-range position, or if two identifiers receive the same position
R: $\mathbf{A} \leftarrow \sum_{i\in I} \dot{\mathbf{x}}_i \dot{\mathbf{x}}_i^\top$, exact and local

Algorithm 3 Upload

for $t = 1, \dots, \bar{n}_O$ **do**
  R: $b_t \leftarrow \mathbb{1}[t \in \pi(I)]$; $\dot{\mathbf{x}}_t \leftarrow \big(1,\ \mathrm{clip}(\mathbf{x}_{R,i}; \sqrt{B_R^2 - 1})\big)$ if $t = \pi(i)$, otherwise $\mathbf{0}$
end for
$m \leftarrow \lceil \bar{n}_O / N_s \rceil$
for each vector $\mathbf{v} \in \{\mathbf{b}, \dot{\mathbf{x}}_{\cdot 0}, \dots, \dot{\mathbf{x}}_{\cdot d_R}\}$ and each chunk $h \le m$ **do**
  R: draw a seed ς; $a \leftarrow \mathrm{PRG}(\varsigma)$; $c_0 \leftarrow -a\, s_R + e + \mathsf{Ecd}_{\Omega_U}(\mathbf{v}[h])$, with e from the RLWE error distribution; send (c_0, ς)
  O: $a \leftarrow \mathrm{PRG}(\varsigma)$; $c_0 \leftarrow c_0 - a\, s_O$ ▷ (c_0, a) now encrypts $\mathbf{v}[h]$ under $s = s_R + s_O$
end for

Theorem 1 (Output privacy against an honest-but-curious R)**.** *Let $\mathcal{P}$ be admissible (Section 7), let every release draw its noise with* SampleDGBudget *at failure bound ξ_T, and let* Calibrate*'s outputs satisfy $\sigma_{c,s} \ge \max\{2^\kappa B_{\mathrm{tot},s}/\sqrt{2\pi}, 2^{42}\}$, $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$ and $\sum_s \rho_s \le \rho_{\mathrm{run}}$, where ρ_{run} converts to $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$ and*

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97} + \xi_T\big), \qquad \delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma > 0.$$

Assume RLWE for the re-randomizer (one sample per release, modulus q_0) and for O's key shares (multi-sample, with circular security for the rotation-key shares; Mouchet et al., 2021), with $\log_2(QP) \le L_{\max}(N, \lambda_{\mathrm{sec}})$. Assume (W): the duration of a fixed sequence of operations of O on fixed-shape operands does not depend on the operand values (machine noise aside). Then R's view of one run, including the time at which each release (or refusal) arrives, is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP for $\sim$ with slack γ, hence $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$-IND-CDP. The ideal mechanism $\mathcal{M}(D) = (f_s(D) + \eta_s)_s$ is ρ_{run}-zCDP. PSI security is not needed: the identifiers are fixed under $\sim$, so the PSI transcript does not depend on $(\mathbf{x}_O, y)$.

Proof sketch. **H0** is the real view. **H0′** replaces each release's noise by the unbudgeted exact sampler's output on the same coins and its timing by that of the sampler's fixed operation sequence: they differ only when the call fails, so this costs ξ_T per release; under (W) the timing is then independent of η_s and of the data. **H1** rewrites $h_{O,s} = v_s - c'_{1,s} s_R$ (identical). **H2** replaces v_s by $f_s(D) + \eta_s$; by Lemma 2 and Obligation 1 this costs at most $\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ per release; with H0′, γ in total. After H2, $(u_s, e_{1,s})$ appear only through $c'_{1,s} = c_{1,s} + a u_s + e_{1,s}$, because $c'_{0,s}$ is never sent. **H3** replaces $c'_{1,s}$ by a uniform element (RLWE with secret u_s, fresh per release). **H4** simulates the setup messages from RLWE on O's key shares. The multivariate discrete Gaussian

Algorithm 4 Aggregate and pack (O)

Input: entry map $e \mapsto (U_e, V_e)$, with R's ciphertext vector U_e and O's plaintext vector V_e in τ-order: $\mathbf{B}_{jk}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{x}_{O,j} \odot \mathbf{x}_{O,k})$; $\mathbf{C}_{jk}$: $(\mathsf{Enc}(\dot{\mathbf{x}}_j),\ \mathbf{x}_{O,k})$; $\mathbf{c}_{R,j}$: $(\mathsf{Enc}(\dot{\mathbf{x}}_j),\ \mathbf{y})$; $\mathbf{c}_{O,k}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{x}_{O,k} \odot \mathbf{y})$; $\mathbf{y}^\top\mathbf{y}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{y} \odot \mathbf{y})$

1: **for** each $e \in \mathcal{E}$ **do**
2: $T_e \leftarrow \textsc{EvalSum}\big(\textsc{Rescale}\big(\sum_{h \le m} U_e[h] \odot \mathsf{Ecd}_{\Omega_V}(V_e[h])\big)\big)$ ▷ sum the chunks first; one EvalSum per entry
3: $P_e \leftarrow \textsc{Rescale}\big(T_e \odot \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_{\mathrm{slot}(e)})\big)$ ▷ $\mathbf{u}_k$ = one-hot slot k; entry e sits only in slot slot(e)
4: **end for**
5: $M_1 \leftarrow \sum_{e \in \mathcal{E}_G} P_e$; $M_2 \leftarrow \sum_{e \in \mathcal{E}_2} P_e$; $M_y \leftarrow P_{\mathbf{y}^\top\mathbf{y}}$ if $mode = \mathsf{separate}$
6: M_1, M_2, M_y are now at q_0 (a modulus switch is a no-op after the two rescales) ▷ depth 2; M_1 depends only on triu $\mathbf{B}$ and $\mathbf{C}$
Scales: R's vectors at Ω_U, masks at Ω_u (parameters), O's data at Ω_V (from Calibrate); $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1q_2)$ is tracked exactly

Algorithm 5 Release(M, σ_c): one DP release

1: **O:** sample fresh u from the key distribution and e_0, e_1 from the error distribution
2: **O:** $c_0' \leftarrow c_0 + p_0u + e_0$; $c_1' \leftarrow c_1 + au + e_1$ ▷ re-randomize $M = (c_0, c_1)$ under the joint pk
3: **O:** $\eta \leftarrow \textsc{SampleDGBudget}(\sigma_c^2, N)$, fresh; **refuse** the release if it fails ▷ exact on $\mathbb{Z}^N$; fixed work; fails w.p. $\le \xi_T$
4: **O:** $h_O \leftarrow c_0' + c_1's_O + \eta \bmod q_0$
5: **O:** send (c_1', h_O); erase c_0', u, e_0, e_1, η ▷ c_0' is never sent; the ledger was charged at admission (implementation, defence in depth: O holds the reply until a public deadline after the request and refuses an overrun)
6: **R:** $v \leftarrow h_O + c_1's_R \bmod q_0$; **return** $\mathsf{Dcd}(v)$ restricted to the slots of this release ▷ exact final scale Ω_f

with sensitivity $\widehat{\Delta}_{\mathrm{enc},s}$ is exactly ρ_s-zCDP (Canonne et al., 2020); costs add; the tight conversion gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP, and SIM-to-IND adds $(1 + e^{\varepsilon_{\mathrm{run}}})\gamma$. □

Remark 1 (Why the release format matters)**.** If O sent c_0', or a share $c_1's_O$ without η, then R would learn $c_1's_O$ exactly and could recover s_O, since c_1' is invertible with overwhelming probability. That was critical issue C3. Sending $h_O = c_0' + c_1's_O + \eta$ and c_1' only means that everything R learns is a function of the noisy value v_s and of c_1', which is pseudorandom. Reusing η or u across releases would reveal $v_2 - v_1$ exactly, so both are fresh every time. Stage 1 uses its own packed ciphertext M_1, which depends only on triu $\mathbf{B}$ and $\mathbf{C}$.

Calibrate. Calibrate (Algorithm 7) derives, from public values only: the mode and n_{rel}; the flooding parameter and the slack; ρ_{run} and its split; the smallest admissible Ω_f (that is, Ω_V) at which every $\sigma_{c,s}^2 = \lceil \widehat{\Delta}_{\mathrm{enc},s}^2/(2\rho_s) \rceil$ floods; and it checks $q_0 > 2(\Omega_f S_{\max} + 2^\kappa B_{\mathrm{tot,max}} + t_\eta \sigma_{c,\max})$ and $\log_2(QP) \le L_{\max}(N, \lambda_{\mathrm{sec}})$ (Homomorphic Encryption Security Standard, 2018). It refuses the run when a step has no solution.

The flooding parameter. The flooding enters the guarantee only through γ, which the SIM-to-IND step multiplies by $1 + e^{\varepsilon}$. Unless $\mathcal{P}$ fixes it,

$$\kappa = \max\{\kappa_{\min}, \kappa_{\mathrm{run}}, \kappa_L\},$$

where κ_{run} is the smallest integer with $(1+e^{\varepsilon_{\text{run}}})\gamma \le \zeta\delta_{\text{run}}$; κ_L is the smallest with $n_{\text{rel}}\varepsilon'(\kappa) \le (\tilde{\rho}_{\text{run}}/\rho_L)E'_L$, the run's share of the ledger's smudging budget, with $\tilde{\rho}_{\text{run}}$ the budget at $\delta_{\text{DP}} = (1-\zeta)\delta_{\text{run}}$ (it holds whenever the zCDP condition does, provided every run uses $\kappa \ge \kappa_L$); and $\kappa_{\min}$ is a statistical security floor. κ_{run} exists iff $n_{\text{rel}}(1+e^{\varepsilon_{\text{run}}})(\delta_B + 2^{-97} + \xi_T) < \zeta\delta_{\text{run}}$. ζ trades budget for bits: it lowers ρ_{run} by a relative $\zeta/\sqrt{L(L+\varepsilon_{\text{run}})}$ to first order, $L = \ln(1/\delta_{\text{run}})$, while each unit of κ costs one bit of Ω_f and of q_0.

Encoding scales. Ω_U and Ω_u are parameters, Ω_V is chosen by CALIBRATE, and Ω_f follows. The term of B_{tot} from R's encryption errors grows like $(\Omega_f/\Omega_U)\sqrt{\bar{n}_O}$; with $\Omega_U \ge \Omega_f$ it does not grow with the register, otherwise the required scale grows with $\bar{n}_O$ and, for large κ, no scale floods.

Ledger and composition across runs

O keeps one ledger per base register, shared across all researchers and scopes; it admits a run only if the run's full cost fits under the cap and produces decryption shares only within admitted runs, so the cap is enforced technically. Its parameters are a cap $(\varepsilon_L, \delta_L)$, a slack share ζ_L, a share ξ_L of ε_L reserved for smudging, and a per-release constant δ' (evaluation: $\xi_L = 10^{-3}$, $\delta' = 2^{-100}$). With $\delta(\rho,\varepsilon)$ the tight conversion, $E'_L = \xi_L\varepsilon_L/2$, $\rho_L = \max\{\rho : \delta(\rho, (1-\xi_L)\varepsilon_L) \le e^{-E'_L}(1-\zeta_L)\delta_L\}$ and $\Gamma_L = \zeta_L\delta_L/(1+e^{\varepsilon_L - E'_L})$, each rounded down. A run costs ρ_{run}, a smudging cost $\varepsilon'_{\text{run}} = n_{\text{rel}}\varepsilon'(\kappa)$ with $\varepsilon'(\kappa) = \min\{\varepsilon : \delta(\pi 4^{-\kappa}, \varepsilon) \le \delta'\} \le 2\sqrt{\pi \ln(1/\delta')}\, 2^{-\kappa} + \pi 4^{-\kappa}$, and a slack $\gamma_L = n_{\text{rel}}(\delta_B + \delta' + \xi_T)$; the state is the three sums $(\rho_\Sigma, E_\Sigma, \Gamma_\Sigma)$ over admitted runs. The full cost of a run, for all its planned releases, is charged atomically at admission and never refunded (refunding the δ_B part of an aborted run would not be sound); requests are processed one at a time. With x^+ the sum after adding the run: runs fixed in advance (parameters and inputs of every run fixed before the first) are admitted iff $e^{E^+}\delta(\rho^+, \varepsilon_L - 2E^+) + (1+e^{\varepsilon_L - E^+})\Gamma^+ \le \delta_L$; runs chosen after seeing earlier results are admitted iff $\rho^+ \le \rho_L$, $E^+ \le E'_L$ and $\Gamma^+ \le \Gamma_L$ (a privacy filter, Rogers, Roth, Ullman and Vadhan, 2016). A single-run register is covered by Theorem 1 alone, with cap $(\varepsilon_{\text{run}}, \delta_{\text{run}})$. The OPRF quota limits membership look-ups across sessions as a matter of policy.

Proposition 1 (Composition across runs)**.** *Let all researchers on a base register act as one PPT coalition, possibly colluding, that chooses each run's parameters and cohort from the joint view of earlier runs, follows the protocol within each run, and may interleave and abort runs. If every admitted run draws fresh key material and release coins and satisfies Theorem 1, the register is static, RLWE holds for every ring parameter set of a finite menu against non-uniform adversaries, and* O*'s ledger is honest, admits runs one at a time and charges a run's full cost at admission, then the joint view of all researchers is $(\varepsilon_L, \delta_L)$-IND-CDP for base-register adjacency (same identifiers and public attributes, one row's $(\mathbf{x}, y)$ changed): under the filter rule for every coalition, and under the first rule when the runs are fixed in advance.*

Proof sketch. Proofs, Part C (Theorems 4.2 and 5.2, Corollary 5.4). One statistical hybrid removes the encryption error from every release of every run, before any computational step; measured in hockey-stick divergence, each release costs $(\varepsilon'(\kappa), \delta')$, and δ_B is charged for every planned release of every admitted run. Computational hybrids then replace each run's c'_1 and key shares run by run. The ideal experiment is ρ_L-zCDP by a Rényi filter proved there. The conversion at $\varepsilon = (1-\xi_L)\varepsilon_L$ gives $(\varepsilon_L, \delta_L)$. Not proved: the first rule with adaptively chosen requests. □

6 Input privacy and cohort privacy

G1. O sees:

- R's uploads $c_0 = -as_R + e + \mathsf{Ecd}(\mathbf{v})$ with a derived from a seed. These are pseudorandom under RLWE in s_R, modelling the PRG as a random oracle.
- The PSI messages, which are BFV ciphertexts under R's own key, and blinded OPRF queries.
- R's key-generation shares, which are RLWE samples.

O never receives a decryption share from R, so its view is simulable from $(n_R^{\max}, \bar{n}_O)$.

G2. O evaluates exactly $n_R^{\max}$ OPRF calls on blinded inputs and answers padded PSI queries, so it learns nothing about the cohort or n_R.

G4. R learns I and $\pi|_I$; that it learns nothing else needs the labeled PSI to be secure against a semi-honest receiver, which Chen, Laine and Rindal (2017) base on IND-CPA security and circuit privacy (statistical function privacy, by re-randomization and noise flooding, suffices). Two layers address the noise of the replies. (a) Every item is $F_k(\mathrm{id}_j)$ under a fresh OPRF key and every label is $\tau(j)$ masked by a key derived from $F'_k(\mathrm{id}_j)$; R learns F_k only at its queries, so items and labels outside I are pseudorandom to it, and a simulator that knows $(I, \pi|_I)$ can rerun O's evaluation on random items and labels outside I: the noise reveals nothing beyond pseudorandom values, under the OPRF's security (OPRF preprocessing, Chen, Huang, Laine and Rindal, 2018). (b) SanitizeBFV is part of the protocol nevertheless: switch the reply and a fresh encryption of zero to the reply modulus q', add them, and flood with $\mathcal{D}_{\mathbb{Z}^{N_B},\sigma_\psi}$, where B_ψ bounds, with probability $\geq 1 - \delta_{B,\psi}$ and for every input of O, the switched reply's error other than the flooding noise and an integral function $\varphi(m)$ of the plaintext with $\|\varphi(m)\|_\infty \leq t/2 + 1$. Requirements: $\sigma_\psi \geq \max\{2^{\kappa_\psi} B_\psi/\sqrt{2\pi}, 2^{42}\}$ and $q'/(2t) > t_\eta \sigma_\psi + B_\psi + t$, the last term covering $\varphi(m)$. Either order of the switch and the addition supports the argument (switching both gives two rounding terms, which B_ψ must cover); this is the implementation's order. Each reply is then within statistical distance $\delta_{B,\psi} + 2^{-\kappa_\psi}(1 + 2^{-80}) + 2^{-97}$, plus an RLWE term, of a ciphertext computed from its plaintext alone. The flooding vectors come from SampleDGBudget, so under (W) the replies' timing adds at most ξ_T per flooding vector ($2\alpha F$ per cuckoo table; Setup checks that the total is at most δ_{psi}) and a failed call aborts the run. Neither layer is charged to δ. A worst-case bound on the BFV evaluation noise inside B_ψ is still open; the implementation's value is measured.

Correctness. With q_0 from Calibrate, honest inputs never wrap around, except with probability at most $\delta_B + 2Ne^{-t_\eta^2/2}$ per release, a correctness event not charged to δ.

7 Parameters

The protocol is defined for a parameter vector $\mathcal{P}$; Table 4 lists each parameter with its constraint and the evaluation setting of the paper and the implementation, which is one admissible choice and not part of the protocol. Calibrate checks the constraints that involve derived quantities and refuses otherwise.

Table 4: Parameters, constraints and evaluation settings.

Parameter	Constraint	Evaluation setting
N, chain q_0, q_1, q_2, P, d_{num}, λ_{sec}	$\log_2(QP) \leq L_{\max}(N, \lambda_{\mathrm{sec}})$ (438 for $N = 2^{14}$ at 128 bits); q_0 above the no-wraparound bound	$N = 2^{14}$; primes 60, 59 (q_0), 59, 59; P two 60-bit primes; $d_{\mathrm{num}} = 2$; $\log_2(QP) = 357$
δ_B	$n_{\mathrm{rel}}(1 + e^{\varepsilon_{\mathrm{run}}})(\delta_B + 2^{-97} + \xi_T) < \zeta\delta_{\mathrm{run}}$; $n_{\mathrm{rel}}(\delta_B + \delta' + \xi_T)$ below each run's ledger share	2^{-50}

Parameter	Constraint	Evaluation setting
ξ_T	failure bound per call of SampleDGBudget, charged per release in γ	2^{-80}
$\kappa_{\min}$; κ	integer floor; κ derived (or fixed with $\delta_{\mathrm{DP}} > 0$)	40; derived
t_η	> 0; correctness failure $2Ne^{-t_\eta^2/2}$ per release	10
$(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$	admitted by the ledger	$\varepsilon \in \{0.5, 1, 2, 4, 8\}$, $\delta = 10^{-9}$
f_1; f_y	$f_1 \in (0,1)$; $f_y \in (0, 1-f_1)$, separate mode only, no default	0.1; 0.1
ζ	$(0,1)$	0.1
$(\varepsilon_L, \delta_L)$, ζ_L, ξ_L, δ', admission rule	public cap per base register, shares in $(0,1)$; filter rule unless runs are fixed in advance; full cost charged at admission	one run per register; example $(10, 10^{-6})$, $\zeta_L = 0.1$, $\xi_L = 10^{-3}$, $\delta' = 2^{-100}$
$\vartheta^\star$; κ_Ψ	$\vartheta^\star > 1$ (> 2 for the refusal bound); $\kappa_\Psi \geq 0$	2.5; 2
$\alpha_{\mathrm{CI}}, \alpha_L, \alpha_U$	positive, sum $\alpha < 1$	0.045, 0.0025, 0.0025
η_B; B	$\eta_B \in (0,1)$; Monte Carlo error $O(B^{-1/2})$	1/2; 2000
B_β^{pub}; bias correction	> 0 or ∞; on or off	∞; off
$\bar{\sigma}^2$	public bound on $\mathrm{Var}(\tilde{y}_i \mid \mathbf{X})$, fixed in advance; $0 < \bar{\sigma}^2 \leq B_y^2$; a value below B_y^2 only as a declared bound on the error variance before clipping when only y is clipped	B_y^2
B_R, B_O, B_y; preprocessing	public, fixed in advance; per-party transforms with public constants	99.5% reference quantiles; standardize, optionally winsorize or block-normalize
Φ, $\bar{n}_O$, $n_R^{\max}$	public attributes only; public sizes	–
N_B, t, h, N_{bins}, d_{cap}, δ_{psi}	t prime, $t \equiv 1 \bmod 2N_B$, $t - 1 \geq n_R^{\max}\bar{n}_O h/(N_{\mathrm{bins}}\delta_{\mathrm{psi}})$	2^{14}, $\approx 2^{32}$ (does not meet the last constraint), 3, 2^{14}, 64, 2^{-40}
κ_ψ, B_ψ, $\delta_{B,\psi}$, q'	flooding and exact-decryption requirements of G4	40; 2^{21} (evaluation noise measured); 0; 2×60 bits
Ω_U; Ω_u; Ω_V, Ω_f	exactly representable; $\Omega_U \geq \Omega_f$ keeps B_{tot} independent of $\bar{n}_O$; Ω_V, Ω_f derived	2^{68}; q_1; derived

Evaluation settings, derived values. For the simulation population ($B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $\Delta = 86.15$, $\Delta_G = 77.16$, joint mode) at $\delta = 10^{-9}$ and a single-run ledger, Calibrate gives $\kappa = 40, 40, 40, 41, 46$ for $\varepsilon = 0.5, 1, 2, 4, 8$, with $\Omega_f = 2^{61.78}$, $2^{62.75}$, $2^{63.69}$, $2^{65.62}$–$2^{65.66}$ and $2^{71.56}$–$2^{71.94}$ for $\bar{n}_O$ from 4,096 to 10^6; q_0 needs at most 98.2 of its 119 bits. At $\varepsilon = 1$: $\sigma_2 \approx 525$, $\sigma_1 \approx 1410$, $B_{\mathrm{tot},2} \approx 2^{26.6}$ with OpenFHE's key switching. Under a ledger cap $(10, 10^{-6})$ with $\zeta_L = 0.1$ and $\xi_L = 10^{-3}$, $\kappa_L = 22, 21, 19, 17, 15$, so $\kappa = 40, 40, 40, 41, 46$ as for a single run (the statistical-distance accounting gave $\kappa_L = 48, 46, 44, 42, 40$ and $\kappa = 48, 46, 44, 42, 46$).

Cost. Table 5 gives the cost per run.

8 Extensions and limitations

A deviating researcher. v3.1 makes no claim against a malicious R. Making G3 hold against one requires:

(i) zero-knowledge proofs at upload that the ciphertexts are well formed with short errors, and that $b_t \in \{0,1\}$, $\|\dot{\mathbf{x}}_t\| \leq B_R$, $(1-b_t)\dot{\mathbf{x}}_t = \mathbf{0}$, $\sum_t b_t \leq n_R^{\max}$, together with a coarse norm bound

Step	Communication	O's computation	Rounds
Link	$n_R^{\max}$ OPRF calls; PSI query and response (sublinear in $\bar{n}_O$)	per-run preprocessing $O(\bar{n}_O)$; per-query evaluation	2–3
Upload	$(d_R + 2)\, m$ seeded ciphertexts (half size)	$(d_R + 2)\, m$ conversions	1
Aggregate	–	$\lvert\mathcal{E}\rvert\, m$ ct–pt products; $\lvert\mathcal{E}\rvert \log_2 N_s$ rotations; $\lvert\mathcal{E}\rvert$ mask products	–
Release (×2 or 3)	2 ring elements each	1 re-randomization, 1 partial decryption, N exact samples	2–3

Table 5: Cost per run, with $m = \lceil \bar{n}_O / N_s \rceil$. O's computation is $\Omega(\bar{n}_O)$, which is inherent for oblivious selection; the public scope Φ is the main lever on it.

that rules out wraparound. Candidates include lattice proofs (LNP22, LaBRADOR) and VOLE-based proofs;

(ii) key generation with R committing first, plus proofs that its shares are short;

(iii) OPRF inputs bound to a certified cohort.

A homomorphic check that relies on R's own decryption share cannot do this job; that was critical issue C1.

Remark 2 (Mode M: hiding membership)**.** Replacing labeled PSI with circuit-PSI (e.g. VOLE-PSI with payloads) gives both parties shares of $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ in τ-order, so R no longer learns I. The aggregation would move to BFV or use wide statistical masking, because shares modulo 2^ℓ do not reduce inside CKKS. $\mathbf{A}$ must then be computed under encryption and noised, and the adjacency extends to changing a record's match status, which raises the sensitivity.

Further limitations.

- Exact-identifier linkage only.
- Linear models only. Generalized linear models via approximate sufficient statistics are future work.
- The variance is model-based. A sandwich variance would need a third release of $\sum_i r_i^2 \mathbf{x}_i \mathbf{x}_i^\top$, a ciphertext–ciphertext computation with its own sensitivity.
- No verifiability of O's computation.

9 Remaining obligations

O1. The worst-case bound B_{tot} (Obligation 1; discharged for the circuit as written).

O2. Smudging for the discrete Gaussian, for every $\sigma_c \geq 2^{42}$ (Lemma 2; proven; the floor 2^{20} stated earlier is too small for the factor $1 + 2^{-80}$), and an exact sampler whose timing does not depend on its output: the fixed-work SampleDGBudget, proved up to its failure bound ξ_T (charged) under the cost assumption (W), which is not verified for the implementation's CPython and numpy operations or for OpenFHE's error sampler.

O3. The simulation-based security of threshold key generation and rotation keys against a semi-honest R, as used in H4.

O4. The validity of the two-stage (λ, Ψ) rule, the RSS correction and the coverage of both intervals, checked by simulation and on real data.

O5. PSI parameters with explicit failure probabilities (correctness only, not charged to δ), and a worst-case bound B_ψ on the evaluation noise of the PSI replies.

O6. Proposition 1: proven, including the adaptive hybrid (proofs, Part C); open only for the first rule with adaptively chosen requests.

Algorithm 6 Solve and infer (R)

Input: $\mathbf{A}$, n, p, slot-domain s.d. $\sigma_1, \sigma_2, \sigma_y$; from $\mathcal{P}$: $\vartheta^\star$, $\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$, B_β^{pub}, the variance bound $\bar{\sigma}^2$, κ_Ψ, bootstrap truncation η_B and size B

$(\hat{\mathbf{B}}_1, \hat{\mathbf{C}}_1) \leftarrow \textsc{Release}(M_1, \sigma_{c,1})$; $\hat{\mathbf{G}}_1 \leftarrow \begin{pmatrix} \mathbf{A} & \hat{\mathbf{C}}_1 \\ \hat{\mathbf{C}}_1^\top & \hat{\mathbf{B}}_1 \end{pmatrix}$

$r_1 \leftarrow \sigma_1\big(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)}\big)$ ▷ $\Pr[\|\mathbf{G} - \hat{\mathbf{G}}_1\|_{\mathrm{op}} > r_1] \le \alpha_L$

$\hat{L} \leftarrow \max\{0,\ \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1\}$; $g^\star \leftarrow 2\vartheta^\star\sigma_2\sqrt{p}$

$\lambda_I \leftarrow \max\{0,\ g^\star + r_1 - \lambda_{\min}(\hat{\mathbf{G}}_1)\}$; $\lambda_O \leftarrow \min\{\lambda \ge 0 : \lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \ge g^\star + r_1\}$ (bisection; ∞ if none)

$(\lambda, \Psi) \leftarrow (\lambda_O, \Pi_O)$ if $\lambda_O \le \kappa_\Psi\lambda_I$, otherwise $(\lambda_I, \mathbf{I})$ ▷ uses stage 1 and $\mathbf{A}$ only

$(\hat{\mathbf{B}}_2, \hat{\mathbf{C}}_2, \hat{\mathbf{c}}_R, \hat{\mathbf{c}}_O, \widehat{yy}) \leftarrow \textsc{Release}(M_2, \sigma_{c,2})$; if separate, $\widehat{yy} \leftarrow \textsc{Release}(M_y, \sigma_{c,y})$

assemble $\hat{\mathbf{G}}_2$ and $\hat{\mathbf{c}}_2$; $\hat{\vartheta} \leftarrow \lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)/(2\sigma_2\sqrt{p})$; **if** $\hat{\vartheta} < 1$ **then return** refused

$\tilde{\mathbf{P}} \leftarrow (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1}$; $\tilde{\boldsymbol{\theta}} \leftarrow \tilde{\mathbf{P}}\hat{\mathbf{c}}_2$ ▷ the bias correction is an option

$\widetilde{\mathrm{RSS}} \leftarrow \widehat{yy} - \hat{\mathbf{c}}_2^\top\tilde{\boldsymbol{\theta}} - \lambda\,\tilde{\boldsymbol{\theta}}^\top\Psi\tilde{\boldsymbol{\theta}}$

for $b = 1, \dots, B$ **do** ▷ parametric bootstrap of the mechanism: post-processing

  draw $\mathbf{E}^*$ (symmetric, s.d. σ_2 on the O-dependent entries, zero on $\mathbf{A}$), $\mathbf{f}^* \sim \mathcal{N}(0, \sigma_2^2\mathbf{I})$, $g^* \sim \mathcal{N}(0, \sigma_{yy}^2)$

  keep the draw only if $\|\tilde{\mathbf{P}}^{1/2}\mathbf{E}^*\tilde{\mathbf{P}}^{1/2}\|_{\mathrm{op}} \le \eta_B$ ▷ truncation: otherwise the moments do not exist

  $\boldsymbol{\theta}_b^* \leftarrow (\hat{\mathbf{G}}_2 + \mathbf{E}^* + \lambda\Psi)^{-1}(\hat{\mathbf{c}}_2 + \mathbf{f}^*)$; $\mathrm{RSS}_b^* \leftarrow$ the same RSS formula at $(\hat{\mathbf{G}}_2 + \mathbf{E}^*, \hat{\mathbf{c}}_2 + \mathbf{f}^*, \widehat{yy} + g^*)$

end for

$\mathbf{V}_{\mathrm{mech}} \leftarrow \widehat{\mathrm{Cov}}_b(\boldsymbol{\theta}_b^*)$; $\mathrm{Corr} \leftarrow \overline{\mathrm{RSS}^*} - \widetilde{\mathrm{RSS}}$; $\tilde{s}^2 \leftarrow \max\{\widetilde{\mathrm{RSS}} - \mathrm{Corr}, 0\}/(n-p)$

CI for the protocol estimand $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$: $\tilde{\theta}_j \pm z_{1-\alpha_{\mathrm{CI}}/2}\sqrt{(\mathbf{V}_{\mathrm{mech}})_{jj}}$

$\mathbf{V} \leftarrow \mathbf{V}_{\mathrm{mech}} + \bar{\sigma}^2\,\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$ ▷ sampling variance from the public bound $\bar{\sigma}^2 \ge \max_i \mathrm{Var}(\tilde{y}_i \mid \mathbf{X})$

$\mathbf{V}^{\mathrm{wm}} \leftarrow \mathbf{V}_{\mathrm{mech}} + \tilde{s}^2\,\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$ ▷ sampling variance under the homoskedastic working model

$U_{yy} \leftarrow \widehat{yy} + z_{1-\alpha_U}\sigma_{yy} + B_y^2\sqrt{n\ln(1/\alpha_U)/2}$; $\hat{B}_\beta \leftarrow \min\{B_\beta^{\mathrm{pub}}, \sqrt{U_{yy}/\hat{L}}\}$ ▷ $\sqrt{\cdot} = \infty$ if $\hat{L} = 0$

$\bar{b}_j \leftarrow \lambda\,\|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\|_2\,\hat{B}_\beta$ ▷ worst-case ridge bias of coordinate j

CI for β_j (bias-aware; Armstrong, Kolesár and Kwon): $\tilde{\theta}_j \pm \mathrm{cv}_{\alpha_{\mathrm{CI}}}\big(\bar{b}_j/\sqrt{V_{jj}}\big)\sqrt{V_{jj}}$ ▷ guaranteed, also when clipping binds

  where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$ quantile of $|Z + t|$, $Z \sim \mathcal{N}(0,1)$

working-model CI for β_j: the same with $\mathbf{V}^{\mathrm{wm}}$ ▷ valid only if the clipped errors are homoskedastic

return $\tilde{\boldsymbol{\theta}}$, the three intervals (each labelled), (λ, Ψ), the shrinkage $-\lambda\tilde{\mathbf{P}}\Psi\tilde{\boldsymbol{\theta}}$, $\hat{\vartheta}$, and (ε, δ) spent

Algorithm 7 Subroutines

function DeltaRep(B_R, B_O, B_y) ▷ replace-one of $(\mathbf{x}_O, y)$, $\dot{\mathbf{x}}$ fixed; $r, o, w = B_R^2, B_O^2, B_y^2$
  if $|r-w| \le 2o$: $\Delta^2 = (2o+r+w)^2/2 + 2rw$; **else if** $r - w \ge 2o$: $\Delta^2 = 4r(o+w)$
  else: one-dimensional maximization, rounded up. Return Δ, and $\Delta_{\neg yy}$ from the same maximization without the $(y^2 - y'^2)^2$ term
  ▷ for $d_O \ge 2$ these are exact suprema; $\Delta_G = \text{DeltaRep}(B_R, B_O, 0) = (2o+r)/\sqrt{2}$ if $r \le 2o$, otherwise $2\sqrt{ro}$
end function
function Calibrate($\mathcal{P}$) ▷ public values only; refuses when a step has no solution
  mode, $n_{\text{rel}} \in \{2, 3\}$, $\{\Delta_s\}$ as in Setup; **refuse if** $\log_2(QP) > L_{\max}(N, \lambda_{\text{sec}})$
  $\gamma(k) := n_{\text{rel}}\big(\delta_B + 2^{-k}(1 + 2^{-80}) + 2^{-97} + \xi_T\big)$; $\varepsilon'(k) := \min\{\varepsilon : \delta(\pi 4^{-k}, \varepsilon) \le \delta'\}$
  unless $\mathcal{P}$ fixes κ: $\kappa \leftarrow \max\{\kappa_{\min}, \kappa_{\text{run}}, \kappa_L\}$ ▷ refuse if none exists
  $\kappa_{\text{run}} = \min\{k : (1 + e^{\varepsilon_{\text{run}}})\gamma(k) \le \zeta\delta_{\text{run}}\}$; $\kappa_L = \min\{k : n_{\text{rel}}\varepsilon'(k) \le (\tilde{\rho}_{\text{run}}/\rho_L)\, E'_L\}$;
  refuse if $n_{\text{rel}}(\delta_B + \delta' + \xi_T) > (\tilde{\rho}_{\text{run}}/\rho_L)\, \Gamma_L$ ▷ ρ_L, E'_L, Γ_L: Section 5
  $\gamma \leftarrow \gamma(\kappa)$; $\delta_{\text{DP}} \leftarrow \delta_{\text{run}} - (1 + e^{\varepsilon_{\text{run}}})\gamma$; **refuse if** $\delta_{\text{DP}} \le 0$
  $\rho_{\text{run}} \leftarrow \max\{\rho : \inf_{a>1} \frac{e^{(a-1)(a\rho - \varepsilon_{\text{run}})}}{a-1}\big(1 - \frac{1}{a}\big)^a \le \delta_{\text{DP}}\}$ ▷ tight zCDP conversion (Canonne et al., 2020)
  $\rho_1 \leftarrow f_1\rho_{\text{run}}$; $\rho_y \leftarrow f_y\rho_{\text{run}}$ (0 in joint mode); $\rho_2 \leftarrow \rho_{\text{run}} - \rho_1 - \rho_y$
  for Ω_f in increasing admissible scales **do** ▷ $\Omega_V = \Omega_f q_1 q_2/(\Omega_U \Omega_u)$, exactly representable
    for each release s: $\widehat{\Delta}_{\text{enc},s} \leftarrow \sqrt{2/N}\,\Omega_f\Delta_s + \Delta_{x,s} + \sqrt{N}$; $\sigma^2_{c,s} \leftarrow \lceil \widehat{\Delta}^2_{\text{enc},s}/(2\rho_s) \rceil$
    if $\sigma_{c,s} \ge \max\{2^\kappa B_{\text{tot},s}(\Omega_f)/\sqrt{2\pi}, 2^{42}\}$ for every s **then break** ▷ worst-case $B_{\text{tot},s}$
  end for ▷ refuse if no scale floods
  $\sigma_s \leftarrow \sigma_{c,s}\sqrt{N/2}/\Omega_f$; **refuse unless** $q_0 > 2\big(\Omega_f S_{\max} + 2^\kappa B_{\text{tot,max}} + t_\eta \sigma_{c,\max}\big)$
  with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$ ▷ no wraparound
  $T_s \leftarrow$ the work budget of SampleDGBudget($\sigma^2_{c,s}, N$) at ξ_T ▷ public; checked by R
  return κ, Ω_V, Ω_f, $\{\sigma^2_{c,s}, \sigma_s, \rho_s, T_s\}_s$, and the run's ledger cost *cost* $= (\rho_{\text{run}},\ n_{\text{rel}}\varepsilon'(\kappa),\ n_{\text{rel}}(\delta_B + \delta' + \xi_T))$
end function
function SampleDGBudget(σ^2, n) n exact samples of $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$, integer σ^2, with a fixed work budget; never by rounding a floating-point normal
  $t \leftarrow 2^b \approx \sigma$; $c \leftarrow \lceil 2^{128}\sigma^2/(2t^2) \rceil / 2^{128}$; $(T, J_v, k_{\max})$ from (σ^2, n, ξ_T) ▷ public
  for $j = 1, \ldots, T$, all of them, with the same fixed-width operations **do** ▷ coins: 128-bit uniforms from a CSPRNG
    $s \leftarrow$ bit; $u \leftarrow \text{U}[0, t)$; $v \leftarrow \#\{1 \le i \le J_v : U < e^{-i}\}$; $y_j \leftarrow u + tv$ if $s = 0$, else $-(u + tv + 1)$
    $g \leftarrow y_j^2/(2\sigma^2) - v + c\ (\ge 0)$; $a_j \leftarrow \text{Bern}(e^{-\lfloor g \rfloor}) \wedge \text{Bern}(e^{-\{g\}})$ ▷ the latter: CKS20 Alg. 1 with exactly $k_{\max}$ coins
  end for
  fail if some trial is undecided ($v = J_v$, all $k_{\max}$ coins 1, or a coin equal to its threshold) or $\sum_j a_j < n$
  return the first n values y_j with $a_j = 1$ ▷ Pr[**fail**] $\le \xi_T$ (Proofs, Part A, Sec. 5.5)
end function
function SanitizeBFV(ct) ▷ statistical flooding of a PSI reply; BFV decryption stays exact
  $ct \leftarrow \mathsf{ModSwitch}_{q'}(ct) \boxplus \mathsf{ModSwitch}_{q'}\big(\mathsf{BFV.Enc}_{pk_B}(\mathbf{0})\big)$; $\mathbf{e} \leftarrow$ SampleDGBudget(σ^2_ψ, N_B) ▷ $\sigma_\psi \ge \max\{2^{\kappa_\psi} B_\psi/\sqrt{2\pi}, 2^{42}\}$
  return $ct \boxplus (\mathbf{e}, 0)$ ▷ requires $q'/(2t) > t_\eta\sigma_\psi + B_\psi + t$
end function
function Preprocess(own columns, D_{ref}) ▷ each party separately; all constants public
  standardize each column with constants from D_{ref} or domain knowledge (O also standardizes y)
  optional tail transform per column: winsorize at $|z| \le c$, rank-Gaussian from the public quantiles, or log
  optional block normalization: scale the party's block of every row to unit norm (R without the intercept)